

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Second Semester of B. Pharm. Examination
University Theory Examination September 2019
BPH2003 Pharmaceutical Engineering

Date: 24/09/2019, Tuesday **Time:** 10:00 a.m. to 01:00 p.m. **Maximum Marks:** 75

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat labeled sketches wherever necessary.
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SECTION – I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. Propeller mixers are operated at _____.
[A] very high speed
[B] very low speed
[C] low speed
[D] medium speed
2. Which of the following is NOT a method used for size reduction?
[A] Cutting
[B] Impact
[C] Burning
[D] Shear
3. Which of the following mill is used for crushing of friable materials?
[A] Cutter Mill
[B] Micronizer
[C] Hammer Mill
[D] Roller Mill
4. Which process uses a porous medium for the separation of the solid material from gas or liquids?
[A] Precipitation
[B] Filtration
[C] Centrifugation
[D] Foam Separation
5. Which is the preferred method of clarification of wine?
[A] Pumping method
[B] Chromatography
[C] Centrifugation
[D] Media storage tank

6. Most commonly used metal for making the sieve is _____.
[A] Zinc
[B] Aluminium
[C] Stainless Steel
[D] Tin
7. Substance exist in more than one crystalline form is called as _____.
[A] polymorphs
[B] amorphous
[C] aggregates
[D] mesostates
8. Which of the following is not a size separation method?
[A] Elutriation
[B] Sieving
[C] Ball mill
[D] Cyclone separator
9. Which of the following is not a type of fire extinguisher?
[A] CO₂
[B] Dry Chemical
[C] Halon
[D] O₂
10. Which of the following method is suitable for grinding of material having water content more than 50%?
[A] Wet grinding
[B] Dry grinding
[C] Cutting
[D] Elutriation
11. The literature of heat transfer generally recognizes distinct modes of heat transfer. How many modes are there?
[A] One
[B] Two
[C] Three
[D] Four
12. Which of the following is an example of forced convection?
[A] Chilling effect of cold wind on a warm body
[B] Flow of water in condenser tubes
[C] Cooling of billets in the atmosphere
[D] Heat exchange on cold and warm pipes
13. Changing of a liquid into vapours from surface of liquid without heating is called as _____.
[A] expansion
[B] contraction
[C] evaporation
[D] fusion
14. Rate of evaporation increases as _____.
[A] the exposed surface area of the liquid increases
[B] the exposed surface area of the liquid decreases
[C] the movement of air above the surface of the liquid decreases
[D] atmospheric pressure increases

15. Which of the following is the first step in making a correct plant location choice?
 - [A] Develop location alternatives
 - [B] Decide the criteria for evaluating location alternatives
 - [C] Evaluate the alternatives
 - [D] Make a decision and select the location
16. Silverson mixer is used for the preparation of _____.
 - [A] Elixir
 - [B] Mouth Wash
 - [C] Emulsion
 - [D] Syrup
17. A jaw crusher uses _____.
 - [A] compression
 - [B] shear
 - [C] cutting
 - [D] attrition
18. The fluid flow in which the fluid particles in one layer do not mix with the fluid particles in the other layer is called as _____.
 - [A] laminar flow
 - [B] turbulent flow
 - [C] layer flow
 - [D] dilatent flow
19. Generally, all the fluid particles in flowing fluid _____.
 - [A] flow at a constant velocity
 - [B] flow at various velocities
 - [C] flow at a velocity as high as possible
 - [D] flow at a velocity as low as possible
20. The moisture inside the substance is known as _____.
 - [A] Bound moisture
 - [B] Unbound moisture
 - [C] Equilibrium moisture
 - [D] Free moisture

SECTION – II

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| Q 2 | Define size reduction. What are the objectives of size reduction? Explain the principle, construction, working and uses of ball mill along with the appropriate diagram. | 10 |
| Q 3 Attempt any <u>TWO</u> of the following | | |
| A | Describe the principle, construction, working and uses of Perforated basket centrifuge along with the appropriate diagram. | 05 |
| B | What is evaporation? Describe the factors affecting evaporation. | 05 |
| C | Explain the principle, construction, working and uses of double cone blender along with the appropriate diagram. | 05 |

SECTION – III

- Q 4** Attempt any **ONE** of the following
- A** What is Filtration? Describe the factors affecting the filtration. Describe the filter aids used in filtration. **10**
- B** What is mixing? What are the objectives of mixing? Describe the mechanisms of mixing along with the factors affecting the mixing process. **10**
- Q 5** Attempt any **FIVE** of the following
- A** Describe objectives and applications of crystallization. **05**
- B** What is heat transfer? Describe the different modes of heat transfer using suitable example. **05**
- C** Explain the principle, construction, working and uses of tray dryer along with the appropriate diagram. **05**
- D** Explain the principle, construction, working and uses of simple distillation along with the appropriate diagram. **05**
- E** What are the different types of industrial hazards? Explain any one hazard in detail. **05**
- F** Describe, in brief, the different types of conveyors with appropriate diagrams. **05**
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